

//SIT 55 : ASSIGNMENT 4 : Dynamic Binding

/*Design a base class Shape with two double type values and member functions to input the data and compute_area() for calculating area of shape. Derive two classes: Triangle and Rectangle. Make compute_area() as abstract function and redefine this function in the derived class to suit their requirements. Write a program that accepts dimensions of triangle/rectangle and display calculated area. Implement dynamic binding for given case study.*/

/*BINDING means connecting a method call to method body.Dynamic Binding takes place at runtime so it is referred as late binding.
Method Overriding is example of Dynamic binding*/

```
package JAVAPROGS;
import java.util.*;
abstract class Shape /*abstract class is a restricted class that
cannot be used to create objects.(to access it, it must be inherited from another class)*/
{
    public double height,width;
    void assign(double h,double w)
    {
        this.height=h;
        this.width=w;
    }
    abstract public void compute_area(); /*abstract method can only be used in an abstract
class, and it does not have a body.
The body is provided by the subclass (inherited from)*/
}

class Triangle extends Shape
{
    public void compute_area()
    {
        System.out.println("AREA OF TRIANGLE="+ (height*width)/2);
    }
}

class Rectangle extends Shape
{
    public void compute_area()
    {
        System.out.println("AREA OF RECTANGLE="+ (height*width));
    }
}

public class Dbinding
{
    public static void main(String[] args)
    {
        float l,b;
        Scanner sc=new Scanner(System.in);
        System.out.println("ENTER LENGTH:");
        l=sc.nextFloat();
        System.out.println("ENTER BREADTH:");
        b=sc.nextFloat();

        Triangle t=new Triangle();
        t.assign(l,b);
    }
}
```

```
t.compute_area();

Rectangle r=new Rectangle();
r.assign(l,b);
r.compute_area();
}
```

```
/*OUTPUT:
ENTER LENGTH:
6
ENTER BREADTH:
5
AREA OF RECTANGLE=30.0
AREA OF TRIANGLE=15.0*/
```